

DSA Practical No: 05

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Div: SY3

Class: C3

Stack Using Array :

INPUT :

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#define SIZE 5
```

```
struct stack
```

```
{
```

```
    int s[SIZE];
```

```
    int top;
```

```
};
```

```
int isFull(struct stack *st)
```

```
{
```

```
    if (st->top >= SIZE - 1)
```

```
        return 1;
```

```
    else
```

```
        return 0;
```

```
}
```

```
int isEmpty(struct stack *st)
```

```
{
```

```
    if (st->top == -1)
```

```
        return 1;
```

```

        else
            return 0;
    }

void push(struct stack *st, int item)
{
    st->top++;
    st->s[st->top] = item;
}

void pop(struct stack *st)
{
    int item;

    if (isEmpty(st))
    {
        printf("\nStack is empty! Underflow.\n");
    }
    else
    {
        item = st->s[st->top];
        st->top--;

        printf("\nThe popped element is: %d\n", item);
    }
}

void display(struct stack *st)

```

```

{
    int i;

    if (isEmpty(st))
    {
        printf("\nStack is empty!\n");
    }
    else
    {
        printf("\nStack elements are:\n");

        for (i = st->top; i >= 0; i--)
        {
            printf("%d\n", st->s[i]);
        }

        printf("TOP = %d\n", st->top);
    }
}

```

```

int main()
{
    struct stack st;
    int choice;
    int item;
    char ans;

    st.top = -1;

```

```

do
{

    printf("IMPLEMENTATION OF STACK\n");
    printf("1. PUSH\n");
    printf("2. POP\n");
    printf("3. DISPLAY\n");
    printf("4. EXIT\n");

    printf("\nEnter your choice: ");
    scanf("%d", Cchoice);

    switch (choice)
    {
        case 1:
            printf("\nEnter the item to be pushed: ");
            scanf("%d", Citem);

            if (isFull(Cst))
            {
                printf("\nStack is full!\n");
            }
            else
            {
                push(Cst, item);
                printf("\nItem %d pushed successfully.\n", item);
            }
    }
}

```

```
break;
```

```
case 2:
```

```
pop(Cst);
```

```
break;
```

```
case 3:
```

```
display(Cst);
```

```
break;
```

```
case 4:
```

```
printf("\nExiting program...\n");
```

```
exit(0);
```

```
default:
```

```
printf("\nInvalid choice!\n");
```

```
}
```

```
printf("\nDo you want to continue? (y/n): ");
```

```
scanf(" %c", Cans);
```

```
} while (ans == 'y' || ans == 'Y');
```

```
printf("\nProgram ended.\n");
```

```
return 0;
```

```
}
```

OUTPUT :

```
PS D:\dsa\Experiment-05-Stack-and-Queue> .\stack.exe
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice: 1

Enter the item to be pushed: 12

Item 12 pushed successfully.

Do you want to continue? (y/n): y
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice: 1

Enter the item to be pushed: 13

Item 13 pushed successfully.
```

```
Do you want to continue? (y/n): y
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice: 1

Enter the item to be pushed: 14

Item 14 pushed successfully.

Do you want to continue? (y/n): y
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice:
1

Enter the item to be pushed: 15

Item 15 pushed successfully.
```

Do you want to continue? (y/n): y

IMPLEMENTATION OF STACK

1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice: 2

The popped element is: 15

Do you want to continue? (y/n): y

IMPLEMENTATION OF STACK

1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice: 3

Stack elements are:

14

13

12

TOP = 2

Do you want to continue? (y/n): y

Do you want to continue? (y/n): y

IMPLEMENTATION OF STACK

1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice: 4

Exiting program...

PS D:\dsa\Experiment-05-Stack-and-Queue> █